

P34 Investor and Customer Brief

Updated release-candidate thesis summary for computable markets

Updated thesis

- P34 turns incomplete business history into executable economic decisions. It is built for markets where the true signal is hidden in operations rather than published in a clean market data feed.
- P34 is a Profit-As-Regression Machine Learning (PARML) model: it trains against final economic outcomes and outputs selected actions and sizes as a portfolio.
- The most important safety action is no-trade: the system should be rewarded for rejecting opportunities when the menu is too uncertain or the expected portfolio result is negative.

What is new in this version

Addition	Why it is worth including
Decision-model framing	Clarifies that P34 is not a price forecaster or dashboard. It is a portfolio construction layer trained against economic outcomes.
Computable-market fit test	Adds practical criteria for deciding whether a market is ready for P34 or needs instrumentation first.
Architecture layers	Explains grounding, regime detection, false-positive correction, universe construction, universe selection, and portfolio execution.
Deeper benchmark evidence	Adds calibration, positive-portfolio rate, oracle efficiency, and benchmark construction metrics beyond headline profit.
Competitive positioning	Separates P34 from AutoML, offline RL, trading platforms, and portfolio analytics tools.
Governance and roadmap	Adds no-trade, audit, kill switches, menu tooling, governance dashboards, and differentiable architecture roadmap.

1. Why this matters now

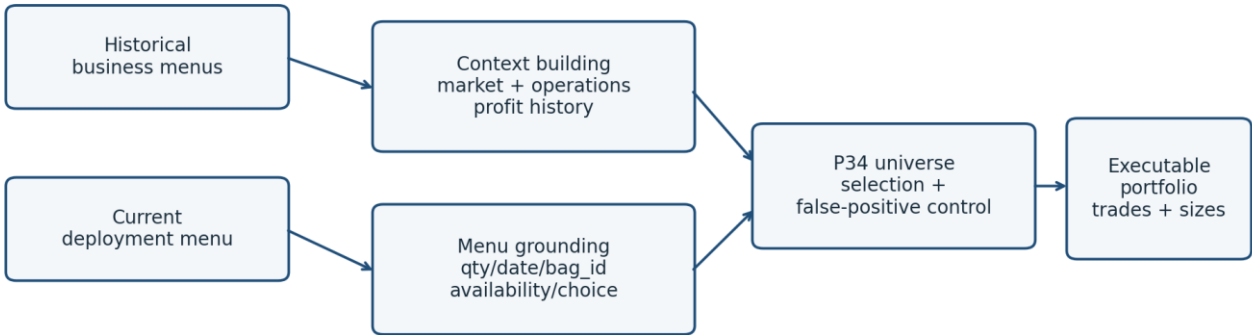
- **Many large markets remain inefficient** because the true market data is not visible. Retail inventory, micro-batch supply, freight capacity, leads, credits, and private-market assets usually lack clean order books or standardized reporting.
- **The economics still exist.** Even when data is not materialized as a Bloomberg Terminal-style feed, it often exists latently inside operations, timing, fulfillment, pricing, routing, inventory movement, and counterparty behavior.
- **Conventional models can become optimistic.** A model trained on historical deals the business already selected may learn from best-effort profitable examples and then over-enter false positives at deployment time.
- **The opportunity is large but not universal.** Retail alone is commonly discussed in the low-tens-of-trillions globally, and the NRF 2026 U.S. retail forecast cited in the source materials is approximately \$5.6T. P34 targets only the computable subset: repeated decisions, measurable outcomes, and enough telemetry for controlled evaluation.

2. What P34 does

P34 evaluates a business menu: a structured set of actions the business could take at a decision point. It learns from historical choices, partial outcomes, safe sizing groundings, and current menus to choose selected actions, sizes, or no-trade.

Concept	Plain-English meaning
Business menu	A table of options the business could choose from. Core fields include qty, date, bag_id, menu_available, and historically_chosen.
Two contexts	A historical context describing market access and business operating performance, plus a deployment-time context describing the current opportunity menu.
Portfolio output	The model selects options and sizes them as an executable portfolio rather than only producing a forecast or dashboard.
False-positive control	The model is designed to learn when not to trade, especially when historical data makes ordinary regressors too optimistic.
Universe selection	P34 constructs multiple plausible interpretations of partial history, filters dominated interpretations, and chooses the one aligned with the policy objective.
Latent-space recovery	The long-term goal is to reconstruct enough hidden market telemetry to support reliable decisions, financing, and liquidity for non-standard assets.

P34 menu-to-portfolio operating concept



Business menu special columns: qty, date, bag_id, menu_available, historically_chosen

Goal: estimate portfolio-wise profit under partial observability and choose the actions worth executing.

Figure 1. P34 converts business menus and context into executable portfolio decisions.

Technical architecture at a glance

Layer	Investor/customer meaning
Menu grounding	Turns operational records into executable options with quantities, dates, menu groupings, availability, choices, and constraints.
Regime detection	Identifies changing market and business states such as margin compression, policy drift, demand shifts, or liquidity changes.
Base profit and risk estimation	Builds nonparametric profit signals and false-positive telemetry from partial labels and safe groundings.
Universe construction	Creates plausible interpretations of the unobserved market instead of pretending the historical data is complete.
Universe selector	Chooses among interpretations using calibration, false-positive control, policy objective, and meta-telemetry.
Portfolio execution	Outputs selected trades and sizes with no-trade available as a valid decision.

3. Proof point from controlled benchmarks

The technical report uses synthetic markets designed to mimic real business constraints: partial observability, selection bias, changing regimes, sparse labels, downside shocks, and optimistic backtest traps. The headline result is not universal profitability; it is evidence that P34 addresses a failure mode common in biased business data.

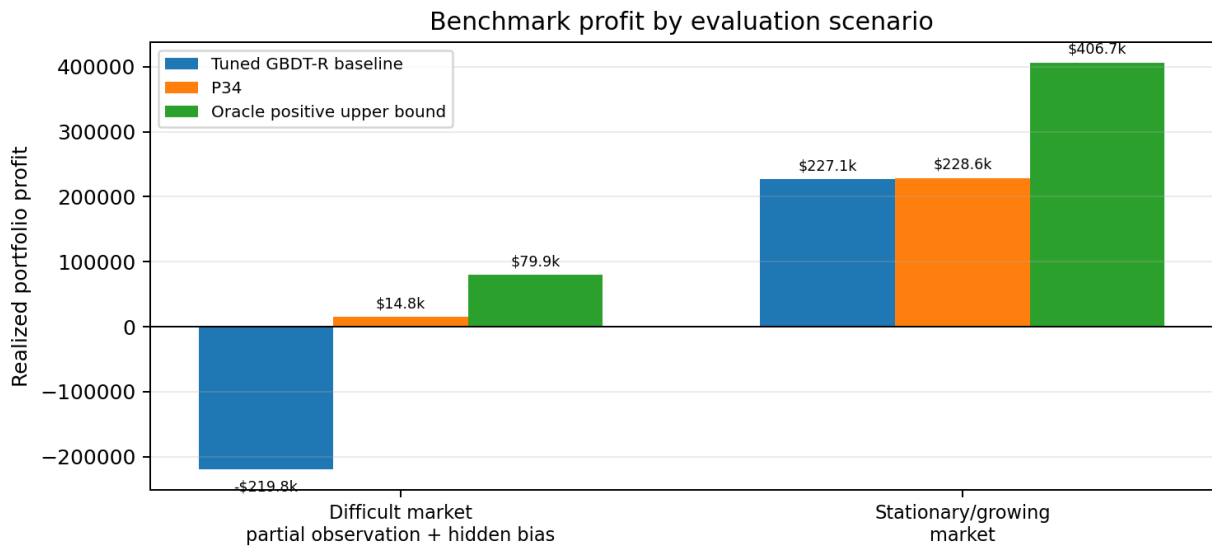


Figure 2. Benchmark results with oracle positive upper bound included for context. The oracle is not deployable.

Metric	Difficult market	Stationary/growing market
GBDT-R predicted profit	+\$126.6k	+\$287.8k
GBDT-R realized profit	-\$219.8k	+\$227.1k
P34 predicted profit	+\$20.7k	+\$378.0k
P34 realized profit	+\$14.8k	+\$228.6k
P34 calibration	71.70%	60.48%
P34 positive-portfolio success	65.00%	90.00%
Oracle positive upper bound	+\$79.9k	+\$406.7k
P34 efficiency vs oracle	18.54%	56.22%

Why the benchmark is a useful stress test

Benchmark property	Inspection value
Telemetry points generated	3,931,200
Historical / future rows	3,024,450 / 906,750
Labeled rows after safe grounding	528,060
Observed-label share after grounding	17.46%
Quantity values per key	39
Positive-profit rate: labeled vs unlabeled	19.23% vs 2.58%
Acceptance-rate variation explained by market + policy	$R^2 = 0.951$ in the diagnostic run

4. Where P34 can apply

The right target is not every market. The right target is a computable market: a market where structured telemetry, repeated decisions, measurable outcomes, and controlled deployment are possible.

Market type	Why P34 may fit
Retail, wholesale, liquidation, apparel, collectibles	Many SKUs, uncertain inventory, discrete quantities, and high value in avoiding bad buys or dead stock.
Freight, load boards, moving leads, call markets	Many candidate bids or leads, partial conversion feedback, and real operational constraints.
GPU capacity, electricity, construction materials, specialty chemicals	Capacity and timing decisions where operational state affects economic value.
Credit, tax credits, environmental credits, short-duration yield markets	Profit depends on risk, timing, adverse selection, recovery, and execution - not a simple prediction.
Private assets and venture investing analogy	A VC sees biased deal flow filtered by networks and prior selection. The same optimistic-data problem can appear in startup investment decisions.

Computable-market fit checklist

Requirement	Practical question
Menu representation	Can available actions be represented as rows with quantities, timing, counterparties, prices, and constraints?
Measurable outcome	Can the business measure profit, contribution margin, recovery, yield, liquidation result, or another economic target after an acceptable delay?
Historical choice signal	Did the business historically have more options than it selected, so prior-policy bias can be estimated?
Executable constraints	Can budget, inventory, compliance, risk, liquidity, capacity, and execution limits be encoded?
Limited or measurable market impact	Does taking one action avoid destroying the validity of the remaining menu in an unobservable way?
Testable deployment path	Can the model be run in backtest, forward test, shadow mode, and capped pilot before scale?

When instrumentation should come first

- If actions cannot be attributed to outcomes, feedback is too delayed or noisy, the menu is not executable, or legal/compliance rules cannot be encoded, the market is not ready for autonomous P34 deployment.
- The first step is telemetry capture, menu design, outcome measurement, and shadow evaluation - not immediate live trading.

5. What customers need to provide

Customer input	Why it matters
Historical business menu	What options were available, what quantities were possible, what the business chose, and what outcomes are known.
Deployment-time menu	The current opportunity set from which P34 should construct a portfolio or decide no-trade.
Economic objective	Profit, contribution margin, recovery rate, yield, utilization-adjusted revenue, or another final business metric.
Historical choices	Prior accepted and rejected options help estimate policy-selection bias rather than treating the data as random.
Safe grounding rules	Examples: smaller quantities may be labelable when a larger executed quantity was observed. These rules expand useful labels without inventing full-market truth.
Operational constraints	Limits on quantity, capital, timing, execution, liquidity, inventory, compliance, risk, and platform rules.
Execution and reconciliation logs	Needed for shadow testing, auditability, leakage checks, calibration, and controlled rollout.

6. Adoption path and controls

Stage	Goal	Evidence required before moving on
1. Ground the menu	Translate business operations into the P34 menu schema.	Schema validation, missingness checks, and execution feasibility review.
2. Train context	Build market-and-business context from historical menus and outcomes.	Leakage checks, date-aware splits, baseline comparison, and calibration diagnostics.
3. Shadow test	Run P34 against live menus without executing its trades.	Compare recommended portfolios with current operations and later realized outcomes.
4. Limited pilot	Execute capped, reviewed, or low-risk portfolios.	Loss limits, exposure caps, manual override, and exception review.
5. Scale with controls	Increase allocation only after evidence of calibration and downside control.	Governance dashboard, kill-switch status, audit logs, drift monitoring, and compliance review.

P34 is not positioned as an uncontrolled trading bot. Deployment should include portfolio caps, review thresholds, drawdown limits, data-quality checks, audit logs, kill switches, allowed-market lists, and market-specific compliance review.

Core control principles

Control	Purpose
No-trade default	Do nothing when expected value is negative or uncertainty is beyond policy limits.
False-positive telemetry	Flag cases that look profitable but historically realize downside.
Drift and regime monitoring	Detect when market conditions differ from historical context.
Independent audit	Review leakage, splits, synthetic generator logic, portfolio reconstruction, and live shadow results.

7. Competitive positioning

P34 does not fit neatly into a single existing software category. It is best understood as a decision layer between enterprise telemetry and executable economic action.

Category	Why it is not the same as P34
AutoML / tabular ML	Can train strong predictors once the target is defined, but does not natively construct business-menu portfolios under partial observation and false-positive risk.
Offline RL / contextual bandits	Conceptually adjacent, but real businesses often lack clean propensities, simulator coverage, or exploration assumptions; P34 emphasizes menu design and conservative universe selection.
Trading and backtesting platforms	Useful for implementing strategies, but usually expect the user to design the strategy. P34 attempts to learn the decision and sizing policy.
Portfolio analytics platforms	Strong for risk, accounting, and compliance workflows, but not publicly positioned as profit-regression engines trained from biased business histories.
Financial LLMs / foundation models	Can provide signals or analysis, but do not themselves output executable menu-based portfolios under business constraints.

8. Current limitations investors and customers should understand

- **Synthetic evidence is bounded.** The principal proof points use synthetic real-world-looking markets; they are stress tests, not live-market performance guarantees.
- **Market computability is not automatic.** P34 does not yet provide a fully automated test proving that a particular market is computable.
- **Menu quality is decisive.** If a business cannot ground actions, quantities, constraints, and outcomes accurately, P34 cannot reliably optimize them.
- **Current implementation is not the final architecture.** Several components use boosted/nonparametric models and materialized matrices today; a more differentiable lifted architecture remains a roadmap item.
- **Latency and unrolling costs matter.** Large menus and portfolio unrolling can create compute costs, and latency may range from under one minute to minutes or hours depending on menu size and configuration.
- **Generalization and adversarial behavior remain open.** Pilots should proceed through shadow testing and controlled deployment, especially in manipulative, low-liquidity, or rapidly changing markets.
- **Distant-menu refusal needs productization.** If a deployment menu is far from training context, an explicit refusal path should be exposed before broad autonomous use.

9. Near-term roadmap

Milestone	Why it matters
Differentiable architecture	Replace or augment boosted blocks with lifted neural components where end-to-end training improves signal flow.
Automated universe scaling	Generate and filter larger sets of plausible market interpretations across broader regime libraries.
Benchmark suite	Create public or auditor-accessible scenarios for declining margin, stationary growth, liquidity shocks, delayed feedback, adversarial players, and sparse labels.
Business menu tooling	Deliver schema validators, menu simulators, leakage tests, and role-specific tooling for business menu managers.
Governance dashboard	Expose regime state, universe choice, false-positive warnings, predicted and realized profit, concentration, and kill-switch status.

Bottom line

- P34 is a new decision layer for partially observed economic markets. Its promise is not only better forecasting, but the ability to recover hidden business signal, reject bad opportunities, and convert large menus of imperfectly observed options into controlled executable portfolios.
- The credible customer path is not immediate full automation. It is menu grounding, context training, shadow testing, capped pilots, and scaling only after calibration and downside-control evidence.

Source notes and publishing caution

Primary source material: HyperC P34 Technical Report draft and benchmark notebooks, June 2026; P34 Industry Article release candidate; P34 Investor and Customer Brief release candidate.

External market-size note: the U.S. retail forecast of approximately \$5.6T for 2026 is attributed in the source materials to National Retail Federation reporting via AP News, March 2026. Refresh external market-size and third-party-source figures before fundraising or customer distribution.

This brief is not investment advice and does not disclose deployment-sensitive architecture, private data, production hardware, training compute, trading endpoints, or live-deployment control logic.